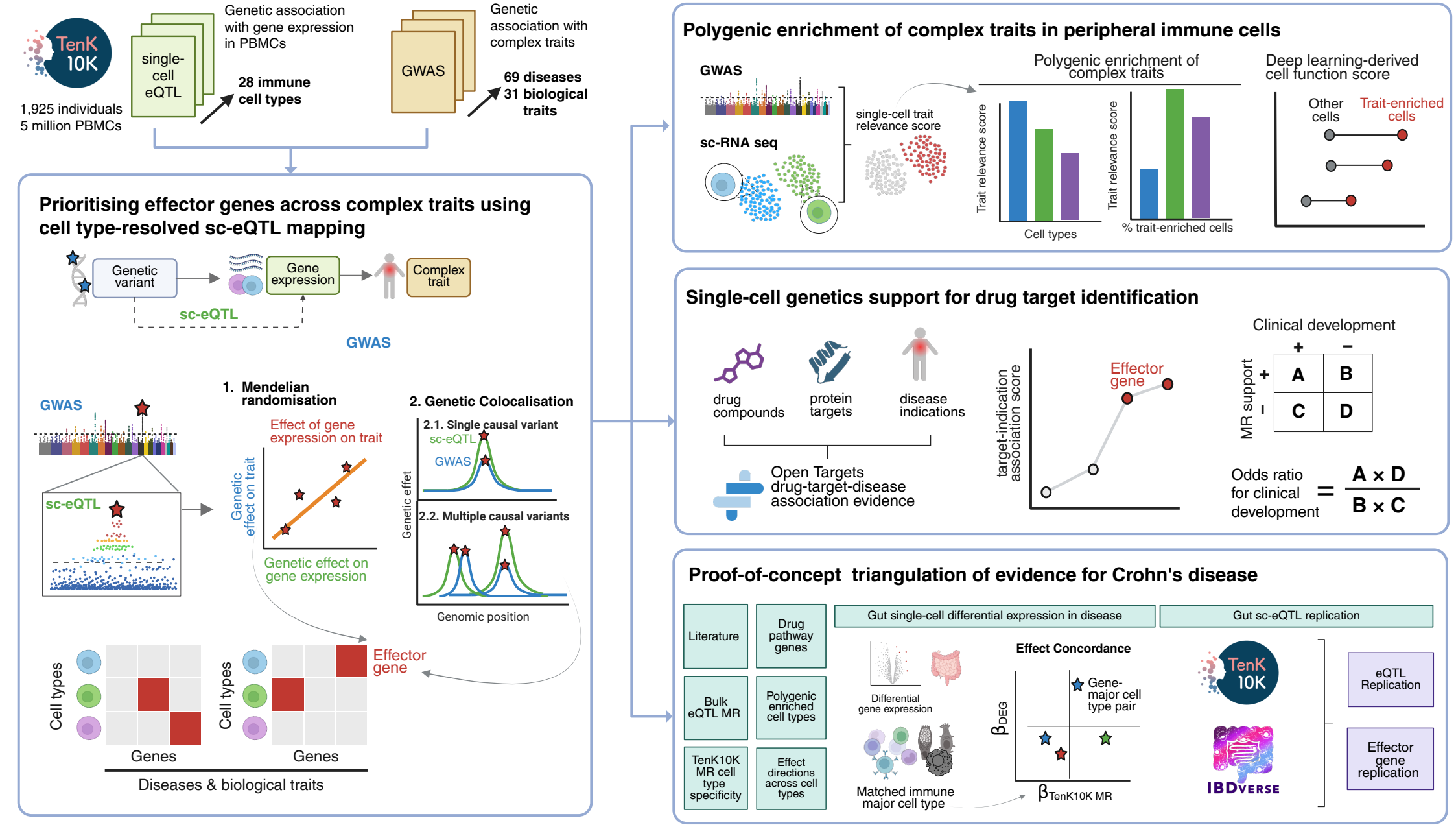

TenK 10K
1,925 individuals
5 million PBMCs
single-cell eQTL
Genetic association with gene expression in PBMCs
28 immune cell types
GWAS
Genetic association with complex traits
69 diseases
31 biological traits
Prioritising effector genes across complex traits using cell type-resolved sc-eQTL mapping
Genetic variant
Gene expression
Complex trait
sc-eQTL
GWAS
GWAS
sc-eQTL
1. Mendelian randomisation
Effect of gene expression on trait
Genetic effect on trait
Genetic effect on gene expression
2. Genetic Colocalisation
2.1. Single causal variant
sc-eQTL
GWAS
2.2. Multiple causal variants
Genetic effet
Genomic position
Effector gene
Cell types
Genes
Cell types
Genes
Diseases & biological traits
Polygenic enrichment of complex traits in peripheral immune cells
GWAS
sc-RNA seq
single-cell trait relevance score
Polygenic enrichment of complex traits
Trait relevance score
Cell types
Trait relevance score
% trait-enriched cells
Deep learning-derived cell function score
Other cells
Trait-enriched cells
Single-cell genetics support for drug target identification
drug compounds
protein targets
disease indications
Open Targets drug-target-disease association evidence
target-indication association score
Effector gene
Clinical development
MR support
+
−
A
B
C
D
Odds ratio for clinical development = A × D / B × C
Proof-of-concept triangulation of evidence for Crohn's disease
Literature
Drug pathway genes
Bulk eQTL MR
Polygenic enriched cell types
TenK10K MR cell type specificity
Effect directions across cell types
Gut single-cell differential expression in disease
Differential gene expression
Matched immune major cell type
Effect Concordance
βDEG
Gene-major cell type pair
βTenK10K MR
Gut sc-eQTL replication
TenK 10K
IBDVERSE
eQTL Replication
Effector gene replication